

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

## SCUT Final Exam

### CALCULUS MID-TERM Exam Paper A (2025-2025-2)

- Notice:
1. Make sure that you have filled the form on the left side of seal line.
  2. Write your answers on the exam paper .
  3. This is a close-book exam.
  4. The exam with full score of 100 points lasts 120 minutes.

| Question No. | I | II | III | IV | V | VI | VII | Sum |
|--------------|---|----|-----|----|---|----|-----|-----|
| Score        |   |    |     |    |   |    |     |     |

1. Answer the questions (16):

- (1) The series as absolutely convergent, conditionally convergent or divergent series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left( \frac{n}{3n+2} \right)^n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^n \frac{\sin n}{n \sqrt{n}} < \frac{1}{n^{\frac{1}{5}}}$$

- (2) Find the least distance between the origin and the surface  $x^2y - z^2 + 9 = 0$

- (3) Find the volume of the tetrahedron with vertices  $(-1, 2, 3)$ ,  $(4, -1, 2)$ ,  $(5, 6, 3)$

- (4) Find the distance between each pair of line  $\frac{x-3}{1} = \frac{y+2}{1} = \frac{z-1}{2}$ ,  $\frac{x+4}{1} = \frac{y+5}{1} = \frac{z}{2}$

- (5) Find  $\nabla f$  and  $D_u f$ , where  $f = x^3 + \tan yz$ ,  $u = (2, 0, -4)$

- (6) The Maclaurin series of  $f(x) = \frac{1}{x^2 + x + 1}$  at  $x = 0$

- (7) The are of one leaf of four leaved rose  $r = 4 \sin 2\theta$

- (8) Evaluate  $\iint_D \frac{x+y}{x-y} dx dy$ ,  $D$  is triangle with vertices  $(1, 0)$ ,  $(4, 0)$ ,  $(4, 3)$

2. Evaluate the problems (30):

- (1) Show the curvature of polar curve  $r^2 = \cos 2\theta$  is directly proportional to  $r$  for  $r > 0$

- (2) Find the convergence set for the power series and find the sum

$$\sum_{n=1}^{\infty} n(n+1)(x-1)^n$$

- (3) Find the equation of the plane through  $(-1, -2, 3)$  and perpendicular to both the

plane  $x - 3y + 2z = 7$  and  $2x - 2y - z = -3$

(4) If  $u(x, t) = \frac{f(x - ct) + f(x + ct)}{2}$ , Show that  $\frac{\partial^2 u(x, t)}{\partial t^2} = c^2 \frac{\partial^2 u(x, t)}{\partial x^2}$

(5) Evaluate  $\int_0^1 \int_0^{\sqrt{1-x^2}} (4 - x^2 - y^2)^{-1/2} dy dx$ .

3. (5) Find the minimum distance from the origin to the line of intersection of two planes

$$x + y + z = 8, \quad 2x - y + 3z = 28$$

4. (5) Evaluate  $\iint_S (x^2 + x^4 y) dA$ . Where  $S = \{(x, y) : 1 \leq x^2 + y^2 \leq 4\}$

5. (5) Switch the order in polar coordinate  $\int_{3\pi/4}^{4\pi/3} d\theta \int_0^{-5\sec\theta} r^3 \sin\theta dr$

6. (7) Find area of region outside the cardioid  $r = 1 + \cos\theta$  and inside the circle  $r = \sqrt{3} \sin\theta$

7. (8) Evaluate the  $\iint_S (x^2 + y^2) dx dy$ . Where  $S$  is bounded by  $9 \leq x^2 + y^2 \leq 16$  and

$1 \leq y^2 - x^2 \leq 9$  in the first quadrant

8. (5) Evaluate  $\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (x^2 + y^2)^{3/2} dy dx$

9 (5) Suppose  $x^3 e^{y+z} - y \sin(x-z) = 0$ , find  $\partial z / \partial x$

10 (4) Rewrite the iterated integral with the indicated order of integration

$$\int_{-1}^0 \int_{-y+1}^{\sqrt{y+1}} f(x, y) dx dy, \quad dy dx$$